

LAB NOC 11

Cisco 888 ISR — DHCP Configuration & Troubleshooting

Real Hardware | ES / EN | NOC Portfolio

INFORMACION DEL LABORATORIO / LAB INFORMATION

Hardware	Cisco Catalyst 2960 + Cisco 888 ISR	Cisco Catalyst 2960 + Cisco 888 ISR
Ticket ID	NOC-LAB-11	NOC-LAB-11
Categoría / Category	DHCP — Configuración y troubleshooting	DHCP — Configuration and troubleshooting
Nivel / Level	CCNA — DHCP / IP Services	CCNA — DHCP / IP Services
Topología / Topology	PC → Switch Fa0/7 → Switch Fa0/16 → Router 888	PC → Switch Fa0/7 → Switch Fa0/16 → Router 888
Pool DHCP	192.168.10.11 – 192.168.10.254 (excluidas .1–.10)	192.168.10.11 – 192.168.10.254 (excluded .1–.10)
Duración / Duration	Aprox. 35 minutos	Approx. 35 minutes
Relevancia NOC	Diagnostico de falla DHCP, APIPA, renovación de lease	DHCP failure diagnosis, APIPA, lease renewal

OBJETIVO / OBJECTIVE

OBJETIVO	OBJECTIVE
Configurar un servidor DHCP en el Cisco 888 ISR para asignar direcciones IP automáticamente a clientes de la red 192.168.10.0/24, simular la falla del servicio DHCP, diagnosticar el síntoma APIPA en la PC, y restaurar el servicio documentando el incidente en formato NOC.	Configure a DHCP server on the Cisco 888 ISR to automatically assign IP addresses to clients on the 192.168.10.0/24 network, simulate DHCP service failure, diagnose the APIPA symptom on the PC, and restore the service documenting the incident in NOC format.

INCIDENT TICKET — NOC-LAB-11

CAMPO / FIELD	DETALLE / DETAIL
Ticket ID	NOC-LAB-11
Title / Título	Usuario sin conectividad de red — PC sin IP / User without network connectivity — PC has no IP
Priority / Prioridad	Medium — usuario productivo sin acceso a red / productive user without network access
Symptom	PC muestra IP 169.254.X.X (APIPA) — no obtiene IP del servidor DHCP / PC shows IP 169.254.X.X (APIPA) — not receiving IP from DHCP server
Root Cause	Servicio DHCP deshabilitado en el router (no service dhcp) / DHCP

Diagnosis Command	service disabled on router (no service dhcp) show running-config include dhcp → confirma 'no service dhcp' activo / confirms 'no service dhcp' active
Resolution	service dhcp en el router + ipconfig /release + ipconfig /renew en la PC / service dhcp on router + ipconfig /release + ipconfig /renew on PC
Validation	ipconfig → IP 192.168.10.X + show ip dhcp binding → MAC de PC con IP asignada
Status	CERRADO — DHCP restaurado, PC con IP correcta / CLOSED — DHCP restored, PC with correct IP

CONFIGURACION DEL POOL DHCP / DHCP POOL CONFIGURATION

PARAMETRO / PARAMETER	VALOR / VALUE	DESCRIPCION / DESCRIPTION
Pool Name	LAN	Nombre del pool DHCP / DHCP pool name
Network	192.168.10.0 255.255.255.0	Red asignada por el pool / Network assigned by pool
Default Router	192.168.10.1	Gateway entregado a los clientes / Gateway delivered to clients
DNS Server	8.8.8.8	Servidor DNS de Google entregado a clientes / Google DNS server delivered to clients
Excluded Range	192.168.10.1 – 192.168.10.10	Reservadas para equipos de red (router, switches) / Reserved for network devices
Assignable Range	192.168.10.11 – 192.168.10.254	Rango disponible para asignacion automatica / Range available for automatic assignment
SVI / Gateway	interface vlan 1 → 192.168.10.1 /24	La SVI del router actua como gateway del pool / Router SVI acts as pool gateway

PROCEDIMIENTO — 5 ETAPAS / PROCEDURE — 5 STAGES

#	ESPAÑOL	ENGLISH
1	ETAPA 1 — Configurar DHCP en el router: ip dhcp excluded-address 192.168.10.1 192.168.10.10	STAGE 1 — Configure DHCP on router: ip dhcp excluded-address 192.168.10.1 192.168.10.10
2	ip dhcp pool LAN → network 192.168.10.0 /24 → default-router 192.168.10.1 → dns-server 8.8.8.8	ip dhcp pool LAN → network 192.168.10.0 /24 → default-router 192.168.10.1 → dns-server 8.8.8.8
3	Configurar SVI: interface vlan 1 → ip address 192.168.10.1 /24 → no shutdown → write memory	Configure SVI: interface vlan 1 → ip address 192.168.10.1 /24 → no shutdown → write memory
4	ETAPA 2 — Configurar PC en modo DHCP automatico (Panel de control → Adaptador → Obtener IP automaticamente)	STAGE 2 — Configure PC in automatic DHCP mode (Control Panel → Adapter → Obtain IP automatically)
5	Verificar: ipconfig → PC recibe 192.168.10.X,	Verify: ipconfig → PC receives 192.168.10.X,

	mask /24, gateway 192.168.10.1	mask /24, gateway 192.168.10.1
6	Verificar en router: show ip dhcp binding → MAC de PC con IP asignada	Verify on router: show ip dhcp binding → PC MAC with assigned IP
7	ETAPA 3 — Simular falla: conf t → no service dhcp → end	STAGE 3 — Simulate failure: conf t → no service dhcp → end
8	Liberar IP de la PC: ipconfig /release → ipconfig /renew → observar IP APIPA 169.254.X.X	Release PC IP: ipconfig /release → ipconfig /renew → observe APIPA IP 169.254.X.X
9	ETAPA 4 — Diagnostico: show running-config include dhcp → confirmar 'no service dhcp'	STAGE 4 — Diagnosis: show running-config include dhcp → confirm 'no service dhcp'
10	ETAPA 5 — Resolucion: conf t → service dhcp → end	STAGE 5 — Resolution: conf t → service dhcp → end
11	En PC: ipconfig /release → ipconfig /renew → confirmar IP 192.168.10.X	On PC: ipconfig /release → ipconfig /renew → confirm IP 192.168.10.X
12	Validar: show ip dhcp binding → MAC presente con IP. Ticket NOC-LAB-11 cerrado.	Validate: show ip dhcp binding → MAC present with IP. Ticket NOC-LAB-11 closed.

CONFIGURACION DHCP PASO A PASO / STEP-BY-STEP DHCP CONFIGURATION

#	COMANDO / COMMAND	ACCION (ES)	ACTION (EN)
1	<code>ip dhcp excluded-address 192.168.10.1 192.168.10.10</code>	Excluir IPs reservadas para equipos de red	Exclude IPs reserved for network devices
2	<code>ip dhcp pool LAN</code>	Crear pool DHCP llamado LAN	Create DHCP pool named LAN
3	<code>network 192.168.10.0 255.255.255.0</code>	Definir la red que el pool va a servir	Define the network the pool will serve
4	<code>default-router 192.168.10.1</code>	Entregar el gateway a los clientes DHCP	Deliver gateway to DHCP clients
5	<code>dns-server 8.8.8.8</code>	Entregar DNS a los clientes	Deliver DNS to clients
6	<code>exit → end</code>	Salir del pool y del modo configuracion	Exit pool and configuration mode
7	<code>interface vlan 1 → ip address 192.168.10.1 /24 → no shutdown</code>	Configurar SVI del router como gateway	Configure router SVI as gateway
8	<code>write memory</code>	Persistir toda la configuracion DHCP	Persist all DHCP configuration

ESCENARIO DE FALLA — DHCP DESHABILITADO / FAILURE SCENARIO

FASE / PHASE	ESPANOL	ENGLISH
Baseline	DHCP activo. PC obtiene 192.168.10.X automaticamente. show ip dhcp binding muestra MAC + IP. / DHCP active. PC automatically gets 192.168.10.X. show ip dhcp binding shows MAC + IP.	DHCP active. PC automatically receives 192.168.10.X. show ip dhcp binding shows MAC + IP.

Falla inducida / Induced failure	conf t → no service dhcp → end (servicio DHCP deshabilitado en el router) / conf t → no service dhcp → end (DHCP service disabled on router)	conf t → no service dhcp → end (DHCP service disabled on router)
Sintoma / Symptom	PC obtiene IP APIPA: 169.254.X.X — indica ausencia de servidor DHCP en la red. Sin conectividad. / PC gets APIPA IP: 169.254.X.X — indicates no DHCP server on network. No connectivity.	PC gets APIPA IP: 169.254.X.X — indicates no DHCP server on network. No connectivity.
Diagnostico NOC / Diagnosis	show running-config include dhcp → muestra 'no service dhcp'. Servicio deshabilitado confirmado. / show running-config include dhcp → shows 'no service dhcp'. Service disabled confirmed.	show running-config include dhcp → shows 'no service dhcp'. Service disabled confirmed.
Resolucion / Resolution	conf t → service dhcp → end. En la PC: ipconfig /release → ipconfig /renew. / conf t → service dhcp → end. On PC: ipconfig /release → ipconfig /renew.	conf t → service dhcp → end. On PC: ipconfig /release → ipconfig /renew.
Validacion / Validation	ipconfig → IP 192.168.10.X (rango correcto). show ip dhcp binding → MAC de PC con IP asignada. Ticket cerrado. / ipconfig → IP 192.168.10.X (correct range). show ip dhcp binding → PC MAC with assigned IP. Ticket closed.	ipconfig → IP 192.168.10.X (correct range). show ip dhcp binding → PC MAC with assigned IP. Ticket closed.

APIPA — DIAGNOSTICO RAPIDO / QUICK DIAGNOSIS

IP / CONCEPTO	NOMBRE / NAME	SIGNIFICADO / MEANING
169.254.X.X	APIPA (Automatic Private IP Addressing)	Windows asigna automaticamente cuando no hay servidor DHCP disponible en la red / Windows automatically assigns when no DHCP server is available on the network
Rango / Range	169.254.0.0 / 16	Rango reservado por IANA para APIPA — no enrutable en Internet / IANA reserved range for APIPA — not routable on Internet
Diagnostico clave / Key diagnosis	PC con 169.254.X.X = sin DHCP	Verificar: (1) router encendido, (2) interfaz up, (3) service dhcp activo / Check: (1) router on, (2) interface up, (3) service dhcp active
En mineria / In mining	IP de camara, PLC o terminal SCADA en 169.254.X.X	Dispositivo OT sin DHCP — verificar pool, interfaz SVI y estado del servicio DHCP antes de escalar / OT device without DHCP — check pool, SVI interface and DHCP service status before escalating

REFERENCIA DE COMANDOS / COMMAND REFERENCE

COMANDO / COMMAND	SIGNIFICADO (ES)	MEANING (EN)
<code>show ip dhcp binding</code>	Muestra todas las IPs asignadas:	Shows all assigned IPs: MAC, IP,

	MAC, IP, expiracion, tipo	expiration, type
<code>show ip dhcp pool</code>	Muestra configuracion del pool: red, gateway, DNS, rango	Shows pool configuration: network, gateway, DNS, range
<code>show ip dhcp server statistics</code>	Estadisticas: mensajes DISCOVER/OFFER/REQUEST/ACK enviados	Statistics: DISCOVER/OFFER/REQUEST/ACK messages sent
<code>show running-config include dhcp</code>	Filtro rapido — muestra si 'no service dhcp' esta activo	Quick filter — shows if 'no service dhcp' is active
<code>service dhcp / no service dhcp</code>	Habilitar / deshabilitar el servicio DHCP en el router	Enable / disable DHCP service on the router
<code>ip dhcp excluded-address X Y</code>	Excluir rango de IPs de la asignacion automatica	Exclude IP range from automatic assignment
<code>ipconfig /release</code>	PC Windows — libera la IP DHCP actual	PC Windows — releases current DHCP IP
<code>ipconfig /renew</code>	PC Windows — solicita nueva IP al servidor DHCP	PC Windows — requests new IP from DHCP server

TROUBLESHOOTING DOCUMENTADO / DOCUMENTED TROUBLESHOOTING

HALLAZGO / FINDING	ACCION CORRECTIVA / CORRECTIVE ACTION
PC obtiene 169.254.X.X (APIPA) en vez de 192.168.10.X — sintoma clasico de servidor DHCP no disponible. / PC gets 169.254.X.X (APIPA) instead of 192.168.10.X — classic symptom of DHCP server unavailable.	Verificar en orden: (1) router encendido y accesible por consola, (2) <code>show ip interface brief</code> — SVI up/up, (3) <code>show running-config include dhcp</code> — confirmar que 'no service dhcp' NO aparece. / Verify in order: (1) router on and accessible via console, (2) <code>show ip interface brief</code> — SVI up/up, (3) <code>show running-config include dhcp</code> — confirm 'no service dhcp' does NOT appear.
La PC obtiene IP DHCP pero esta fuera del rango esperado — por ejemplo obtiene una IP que deberia estar excluida. / PC gets DHCP IP but outside expected range — for example gets an IP that should be excluded.	Verificar con <code>show ip dhcp pool</code> la lista de exclusiones. Si falta el <code>ip dhcp excluded-address</code> , los equipos de red pueden recibir IPs que ya estan asignadas estaticamente al router o switches, generando conflictos. / Verify with <code>show ip dhcp pool</code> the exclusion list. If <code>ip dhcp excluded-address</code> is missing, network devices may receive IPs already statically assigned to routers or switches, causing conflicts.
Despues de <code>service dhcp</code> la PC aun muestra 169.254.X.X aunque el servidor esta activo. / After <code>service dhcp</code> the PC still shows 169.254.X.X even though server is active.	El cliente DHCP de Windows necesita liberar y renovar explicitamente: <code>ipconfig /release</code> seguido de <code>ipconfig /renew</code> . Sin este paso la PC no solicita nueva IP aunque el servidor este disponible. / Windows DHCP client needs explicit release and renew: <code>ipconfig /release</code> followed by <code>ipconfig /renew</code> . Without this step the PC does not request a new IP even if the server is available.

RESULTADO / RESULT**RESULTADO (ES)**

DHCP configurado y operativo en Cisco 888. Pool LAN asignando 192.168.10.11-254. Falla simulada (no service dhcp) diagnosticada via APIPA. Servicio restaurado con service dhcp + ipconfig /renew. Ticket NOC-LAB-11 cerrado.

RESULT (EN)

DHCP configured and operational on Cisco 888. LAN pool assigning 192.168.10.11-254. Simulated failure (no service dhcp) diagnosed via APIPA. Service restored with service dhcp + ipconfig /renew. Ticket NOC-LAB-11 closed.

KEY TAKEAWAY

"En NOC, el primer indicador de falla DHCP es una IP 169.254.X.X (APIPA) en el cliente. El diagnostico es inmediato: show running-config | include dhcp en el router. Si aparece 'no service dhcp', un solo comando lo resuelve. El error mas comun post-resolucion es olvidar el ipconfig /renew en la PC — el servidor puede estar activo pero el cliente Windows no solicita nueva IP hasta que se le indica explicitamente."

"In NOC, the first indicator of DHCP failure is a 169.254.X.X (APIPA) IP on the client. Diagnosis is immediate: show running-config | include dhcp on the router. If 'no service dhcp' appears, a single command resolves it. The most common post-resolution error is forgetting ipconfig /renew on the PC — the server may be active but the Windows client does not request a new IP until explicitly told to."

NOC RELEVANCE — MINING / INDUSTRIAL ENVIRONMENT

In a mining operation (e.g., Newmont Cerro Negro), DHCP failures are high-frequency tickets — particularly after router reloads, maintenance windows, or configuration changes that accidentally disable the service. A NOC engineer must recognize APIPA (169.254.X.X) as an immediate DHCP failure signature, know the three-command diagnosis sequence (show ip dhcp binding → show ip dhcp pool → show running-config | include dhcp), and understand that OT devices (IP cameras, PLCs, SCADA terminals) configured for DHCP will lose connectivity silently if the service is disabled. For critical OT devices, static IP assignment with DHCP exclusion of those addresses is the recommended practice to avoid accidental IP conflicts.